

## Mid-Range Graphic Cards

|                                                                  |                                                                                             |                                                                                             |                                                                     |                                    | Above Rs. 10,000                                                                                                |                                                                                        |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Brand                                                            | MSI                                                                                         | MSI                                                                                         | ZOTAC                                                               | ATI                                | ASUS                                                                                                            | Sapphire                                                                               |
| Model                                                            | R4850-T2D512                                                                                | N9800GTX-T2D512                                                                             | GeForce GTS 250 AMP! 1 GB                                           | Radeon HD 5770                     | EN 9800 GT HB                                                                                                   | Radeon HD 4850 1 GB                                                                    |
| GPU                                                              | Radeon HD 4850                                                                              | GeForce 9800GTX                                                                             | GeForce GTS 250                                                     | Radeon HD 5770                     | GeForce 9800GT                                                                                                  | Radeon HD 4850                                                                         |
| Price (Rs)                                                       | Rs. 9,000                                                                                   | Rs. 9,500                                                                                   | Rs. 9,900                                                           | Rs. 9,990                          | Rs. 10,850                                                                                                      | Rs. 10,900                                                                             |
| Grand Totals (Out of 100)                                        | 56.93                                                                                       | 59.74                                                                                       | 69.36                                                               | 72.31                              | 51.61                                                                                                           | 63.77                                                                                  |
| Features (Out of 15)                                             | 6.98                                                                                        | 5.85                                                                                        | 8.18                                                                | 7.13                               | 6.98                                                                                                            | 6.38                                                                                   |
| Performance (Out of 85)                                          | 49.95                                                                                       | 53.89                                                                                       | 61.18                                                               | 65.18                              | 44.63                                                                                                           | 57.39                                                                                  |
| Features                                                         |                                                                                             |                                                                                             |                                                                     |                                    |                                                                                                                 |                                                                                        |
| No. of Stream Processors                                         | 160 (800 / 5)                                                                               | 128                                                                                         | 128                                                                 | 800                                | 112                                                                                                             | 160 (800 / 5)                                                                          |
| Core / Shader Clock (MHz)                                        | 625 / -                                                                                     | 720 / 1750                                                                                  | 750 / 1890                                                          | 850 / -                            | 600 / 1500                                                                                                      | 625 / -                                                                                |
| Memory Type / Memory Clock (MHz)                                 | GDDR3 / 1986                                                                                | GDDR3 / 2200                                                                                | GDDR3 / 2300                                                        | GDDR5 / 4800                       | GDDR3 / 1800                                                                                                    | GDDR3 / 1986                                                                           |
| Memory size (MB) / Interface                                     | 512 / 256-bit                                                                               | 512 / 256-bit                                                                               | 1024 / 256-bit                                                      | 1024 / 128-bit                     | 512 / 256-bit                                                                                                   | 1024 / 256-bit                                                                         |
| Connects on card                                                 | 2 x DVI, 1 x S-Video                                                                        | 2 x DVI, 1 x S-Video                                                                        | 2 x DVI, 1 x S-Video                                                | 2 x DVI, 1 x HDMI, 1 x DisplayPort | 2 x DVI, 1 x S-Video                                                                                            | 1 x DVI, 1 D-Sub, 1 x HDMI                                                             |
| In the Box                                                       |                                                                                             |                                                                                             |                                                                     |                                    |                                                                                                                 |                                                                                        |
| Bundled Connects and Cables                                      | 1 x DVI-D-Sub converter, Crossfire connect, S-Video-Composite connect, S-Video to component | 1 x DVI-D-Sub converter, Crossfire connect, S-Video-Composite connect, S-Video to component | 1 x DVI - D-Sub converter, 2 x 6 pin to 4 pin molex power converter | -                                  | 1 x DVI - D-Sub converter, 1 x DVI - HDMI converter, HDMI audio cable, 1 x 6 pin to 4 pin molex power converter | 1 x DVI - D-Sub converter, 1 x HDMI converter, 1 x 4 pin - 6 pin molex power converter |
| Software and Games                                               | -                                                                                           | -                                                                                           | XIII Century: Death Or Glory                                        | -                                  | -                                                                                                               | -                                                                                      |
| Performance                                                      |                                                                                             |                                                                                             |                                                                     |                                    |                                                                                                                 |                                                                                        |
| Temperature (Idle / load)                                        | 45 / 85                                                                                     | 40 / 58                                                                                     | 44 / 57                                                             | 46 / 57                            | 52 / 62                                                                                                         | 44 / 56                                                                                |
| 3D Mark Vantage (Overall / GPU)                                  | 8322 / 6936                                                                                 | 8223 / 6462                                                                                 | 8699 / 6853                                                         | 10124 / 8657                       | 6019 / 4623                                                                                                     | 8480 / 7077                                                                            |
| Game Benchmarks (in frames per second)                           |                                                                                             |                                                                                             |                                                                     |                                    |                                                                                                                 |                                                                                        |
| Crysis Warhead (1680 x 1050, Max / Avg / Min)                    | 19 / 14 / 9                                                                                 | 21 / 14 / 8                                                                                 | 23 / 16 / 10                                                        | 22 / 16 / 10                       | 18 / 12 / 7                                                                                                     | 29 / 16.7 / 10.2                                                                       |
| Crysis Warhead (1280 x 1024, Max / Avg / Min)                    | 25 / 17 / 12                                                                                | 29 / 18 / 11                                                                                | 30 / 20 / 13                                                        | 28 / 19 / 13                       | 23 / 15 / 9                                                                                                     | 40 / 21.4 / 12.4                                                                       |
| S.T.A.L.K.E.R Clear Sky (Day, 1680 x 1050, Max / Avg / Min)      | 28.5 / 15.5 / 9                                                                             | 41.3 / 13.5 / 5.3                                                                           | 31.5 / 16.8 / 10.2                                                  | 36.9 / 19.3 / 10.5                 | 23.3 / 9.9 / 4.5                                                                                                | 29 / 16.9 / 10.6                                                                       |
| S.T.A.L.K.E.R Clear Sky (Day, 1280 x 1024, Max / Avg / Min)      | 39.6 / 20.1 / 7.5                                                                           | 44.3 / 17.2 / 7.5                                                                           | 46.7 / 21.4 / 12.9                                                  | 50.3 / 23.9 / 13.5                 | 34.2 / 12.3 / 2.6                                                                                               | 40.1 / 21.4 / 12.3                                                                     |
| Far Cry 2 (1680 x 1050, Ultra High, 4x AA) Max / Avg / Min       | 37.8 / 18.06 / 7.81                                                                         | 56.33 / 34.23 / 11.3                                                                        | 58.8 / 41.41 / 30.13                                                | 150.41 / 43.45 / 28.02             | 43.23 / 26.23 / 16.85                                                                                           | 50.27 / 33.9 / 23.14                                                                   |
| Far Cry 2 (1280 x 1024, Ultra High, 4x AA) Max / Avg / Min       | 53.57 / 36.99 / 24.03                                                                       | 65.27 / 41.62 / 18.78                                                                       | 80.36 / 47.23 / 35.32                                               | 130.49 / 51.4 / 36.82              | 46.32 / 32.12 / 24.78                                                                                           | 149.5 / 40.56 / 28.73                                                                  |
| Unreal Tournament 3 (1680 x 1050 / 1280 x 1024)                  | 107 / 133                                                                                   | 128 / 146                                                                                   | 129 / 152                                                           | 125 / 153                          | 93 / 112                                                                                                        | 107 / 135                                                                              |
| Company Of Heroes: Tales Of Valor (1680 x 1050, Max / Avg / Min) | 195 / 101.8 / 47.1                                                                          | 323.2 / 137.4 / 61.4                                                                        | 289.1 / 138.5 / 67.5                                                | 282.8 / 147.1 / 69.5               | 183.3 / 89.3 / 43.2                                                                                             | 188 / 102.5 / 45.3                                                                     |
| Company Of Heroes: Tales Of Valor (1280 x 1024, Max / Avg / Min) | 246.2 / 135.8 / 69.7                                                                        | 412 / 176.1 / 89.2                                                                          | 398.8 / 197.2 / 97.3                                                | 383 / 200.3 / 106.4                | 302.5 / 129.4 / 64.9                                                                                            | 237 / 136.9 / 70                                                                       |
| Visual Quality (Out of 10)                                       |                                                                                             |                                                                                             |                                                                     |                                    |                                                                                                                 |                                                                                        |
| Crysis Warhead / S.T.A.L.K.E.R Clear Sky                         | 8 / 8                                                                                       | 8 / 7.75                                                                                    | 8 / 7.75                                                            | 9 / 8.5                            | 8 / 7.75                                                                                                        | 8 / 8                                                                                  |

and not the price tag of the card that determines its placement either the high-end or mid-range category. ASUS' Radeon HD 4850 MATRIX has a dual-slot, dual-fan cooler but it's the tiny cooler aboard the ASUS 4850 TOP that takes the cake as far as cooling goes. MSIs N9800GTX PLUS-T2D512-OC was a real cool customer in 2D mode, but start up a game and this card clocked the highest temperature in this category (65 degrees). While not dangerous, you will need good air circulation inside the cabinet you're using.

Among these cards as in

the previous category, we see the Radeon HD 4850 vying for power with the GeForce 9800 GTX cards. With the resolution and details cranked all the way up, these cards start to huff and puff when running *Clear Sky* and *Warhead*. At times we see an inexplicable drop in frames, like the ASUS Radeon HD 4850 TOP dropping to a minimum of 4 fps in Far Cry 2 at a resolution of 1680 x 1050. Such oddities are difficult to explain – it could be due to the card overheating at some point or some internal fault in terms of design could be the culprit. The ATI cards seem

a little better equipped to handle demanding resolutions and the Sapphire HD 4850 with 1 GB of RAM clearly has a lead especially in the minimum fps values. This makes us believe that the extra video RAM is alleviating the bottleneck that occurs with higher resolutions and visual settings. This means the GPU on the 4850 is powerful enough to be bottlenecked by 512 MB of RAM, something not commonly seen on cards in this price range. Unfortunately, this card is also comparable in price to some of the high-end graphics cards including some of the cheaper

Radeon HD 4870 cards, needless to say someone shopping for a graphics card in this price range should consider buying one based on a higher GPU.

## High-end cards up to Rs. 17,500

A small but increasing number of gamers and enthusiasts in our country are shopping for the right GPU in the price range of Rs. 15,000 and though, (we repeat), this is a rarity it's also an interesting category for a number of reasons. For one, prices of GPUs have tumbled and while Rs. 15,000 may be considered